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ORIGINAL ARTICLE

A new exploration in Baltic Dry Index forecasting learning: application of a deep ensemble model

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Abstract

World trade is growing constantly, facilitated by the fast expansion of logistics. However, risks and uncertainty in shipping have also increased, in dire need to be addressed by the research community, through more accurate and efficient methods of forecasting. In recent years, combining attention models and deep learning has produced remarkable results in various domains. With daily data spanning the period from January 6, 1995, to September 16, 2022 (totaling 6896 observations), we predict the Baltic Dry Index (BDI) using a deep integrated model (CNN-BiLSTM-AM) comprising a convolutional neural network (CNN), bi-directional long short-term memory (BiLSTM), and the attention mechanism (AM). Our findings indicate that the integrated model CNN-BiLSTM-AM encompasses the nonlinear and nonstationary characteristics of the shipping industry, and it has a greater prediction accuracy than any single model, with an R^2 value of 96.9%. This research shows that focusing on the data's value has a particular appeal in the intelligence era. The study enhances the integrated research of machine learning in the shipping business and offers a foundation for economic decisions.

Keywords Logistics · Shipping business · Forecasting · Baltic Dry Index · Machine learning · Shipping economics

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1 Introduction

The shipping sector has made a vital contribution to the economic growth of many countries and it plays an essential part in the world trade system (Kanamoto et al. 2021). Shipping transports 80% to 90% of international trade volume and is thus one of the most important transport modes in the world (Zhang et al. 2019). The industry's volatility indicators are composite economic indicators (Duru et al. 2012; Chang et al. 2014). Among the many, the Baltic Dry Index (BDI), published by the *Baltic Exchange*, is a key price (freight rate) indicator (Cullinane et al. 1999; Lin et al. 2019). BDI is often considered as the "barometer" of the shipping industry, as well as a leading indicator of general economic activity (Liu et al. 2022). Specifically, the BDI analyzes transportation prices for dry bulk cargo such as iron ore, coal, and grains from producing countries such as Brazil, Australia, and South Africa to markets in the United States, Europe, Japan, and China (Duru et al. 2012).

As a benchmark for measuring transportation costs, researchers have often considered the BDI as an ideal proxy variable to global economic activity. Changes in BDI have been seen as a leading measure of future economic growth or contraction (Lin and Sim 2013; Gu et al. 2019; Haralambides 2019). In finance, investment bankers and hedge fund managers commonly view the BDI as a leading economic indicator for financial markets such as equities and commodities (Alizadeh and Muradoglu 2014). The BDI is also a crucial indicator for bulk carrier owners, charterers, and shippers when making crucial decisions or implementing strategic business plans (Chou and Lin 2019; Yang and Mehmed 2019), because the index can assist stakeholders with maritime business planning and market risk avoidance. Therefore, an accurate prediction of the BDI provides a decision basis for decision-makers in the shipping sector, industry, finance, and government agencies and it is beneficial to the healthy growth of the global economic system (Chen et al. 2012; Apergis and Payne 2013; Wang et al. 2015).

In the past 40 years, the BDI has been an important research area (Duru and Yoshida 2009). There are three primary aspects in BDI research. First, BDI is used by researchers as an indirect indicator of market expectations regarding the global economy and trade. Authors use it as a predictor of various economic variables such as exchange rate fluctuations, inflation, and demand for transport of raw materials, energy, and agricultural products (Bildirici et al. 2015; Papailias et al. 2017; Han et al. 2020). The quantitative relationship between the BDI and commodity markets constitutes the second area of study (Lin et al. 2019; Said and Giouvris 2019; Bandyopadhyay and Rajib 2023). The third aspect concerns BDI forecasting.

The BDI is a typical nonstationary nonlinear time series influenced by numerous complex variables (Liu et al. 2022). Although the literature provides various studies and analyses of BDI trends, the time-varying and unpredictable character of the shipping industry makes it difficult, if not impossible, to predict it with a satisfactory degree of accuracy. Freight rate forecasting is unquestionably an essential albeit difficult endeavor (Liu et al. 2022).

In summary, the logistics industry urgently requires effective data mining techniques. For example, earlier BDI prediction research used predominantly



causality-based econometric models which, however, required the regularity and stationarity of large datasets (Duru 2010; Duru et al. 2012; Gao et al. 2023). Nonetheless, the data features of the BDI cause significant mistakes in the precision of econometric models (Zhang et al. 2018). However, research has demonstrated that AI-based methods yield many superior outcomes compared to conventional econometric and statistical models. Consequently, researchers have utilized numerous nonlinear and artificial intelligence (AI) techniques, such as artificial neural networks (ANN), support vector machine (SVM), and nonlinear regression (Yang et al. 2008; Chou and Lin 2019). However, the limited nature of a single AI prediction approach makes it susceptible to over- or undertraining, resulting in errors. Research has thus questioned a single machine learning method (Zhang et al. 2019). In addition, most previous studies viewed BDI prediction as a forecast of a single-day variable (Kamal et al. 2020). But markets need projections for at least one week in advance, given the volatility of freight rates (Bae et al. 2021). In conclusion, there are still significant discrepancies in the BDI research, particularly in BDI prediction.

This study, therefore, collected daily prices from January 6, 1995, to September 16, 2022, totaling 6896 observations. We employ an integrated model (CNN-BiLSTM-AM) comprising a convolutional neural network (CNN); bi-directional long short-term memory (BiLSTM); and attention mechanism (AM), to increase the accuracy of BDI freight rate forecasting.

Among the many contributions of this study is our novel approach based on deep learning ensemble models that effectively capture the nonlinear and nonstationary properties of freight rates. This study also expands the scope of forecasting methodologies. Second, BDI forecasting studies often include daily forecasts. This study increases the forecasting horizon and enhances the forecasting cycle, maintaining forecast accuracy.

The remainder of this study is as follows. Section 2 reviews the methods used to forecast BDI time series, machine learning, and forecasting combined methods. Section 3 presents our data and the ensemble methods used here. Section 4 discusses the main empirical findings and compares them with alternative models. Finally, Sect. 5 discusses the research results, concludes, and suggests further research avenues.

2 Literature review

2.1 Econometric models

Freight rate modeling and forecasting research has taken place for more than half a century (Duru 2010). The most prevalent methods have been econometric modeling or machine learning, which include time series, linear regression, fuzzy sets, and artificial neural networks (Zhang et al. 2018; Chou and Lin 2019). There is widespread usage of linear regression models for predictive analysis in the context of price volatility (Bae et al. 2021). Models of autoregressive integrated moving average (ARIMA), vector autoregressive (VAR), generalized autoregressive conditional heteroscedasticity (GARCH), and vector error correction (VEC) are examples of



econometric techniques (Bildirici et al. 2015; Zhang et al. 2019; Xiao et al. 2021). Initially, a study developed a VAR model for predicting the BDI using a coin integration method and a unit root test for the time series (Veenstra and Franses 1997). Another study introduced the ARIMA model to dry bulk freight rate forecasting for the first time, to forecast the impact of freight rates on the shipping industry (Cullinane et al. 1999). Academics have readily understood the significance of applying several econometric models to freight rate forecasting.

Using autoregressive integral moving average and VAR models, Kavussanos and Alizadeh (2001) examined the seasonal characteristics of the dry bulk shipping industry. However, another study using ARIMA and VAR to anticipate freight rates for multiple dry bulk routes indicated that VAR outperforms ARIMA in out-of-sample forecasting (Chen et al. 2012). In addition, Tsioumas et al. (2017) validated the initial research findings. They propose a multivariate VAR model (VARX) using exogenous variables and discover that the VAR model outperforms the ARIMA technique. In addition, these researchers discussed the performance of VAR, ARIMA, and vector balance correction (VECM) approaches for predicting spot and forward freight rates. They discovered that VAR and VECM predicted spot freight rates somewhat better than ARIMA (Batchelor et al. 2007). Moreover, Kavussanos and Alizadeh (2001) discovered that the VECM model predicts freight rates more accurately than the ARIMA model.

The majority of these classic statistical analyses include error measures such as the mean absolute percentage error (MAPE), the mean square error (MSE), and the root mean square error (RMSE) to improve the prediction accuracy. Nonetheless, such conventional methods produce many inconsistent results because of the shortcomings of statistical extrapolation. Specifically, linear regression forecasting models, based on mathematical equations, cannot learn dynamically; predict patterns of change; or identify moving trends (Lin et al. 2007). In addition, traditional linear regression models and time series methods are characterized by data intensity, noise, non-stationarity, high uncertainty, and implied relationships in the actual forecasting process. This can reduce the accuracy and stability of model predictions (Chou and Lin 2019). Therefore, research has gradually shifted focus to AI techniques to investigate the inherent complexity of the BDI (Zhang et al. 2019).

2.2 Machine learning models

Freight market forecasting outcomes have had a poor reputation before machine learning advances, due to the difficulty of capturing the hidden nonlinear properties of the dry bulk freight index with conventional econometric methods (Zhang et al. 2018, 2019; Katris and Kavussanos 2021). As a result, academics have increasingly focused on various AI techniques. The SVM, ANN nonlinear regression and fuzzy system are excellent machine learning-based prediction approaches that have emerged in recent years (Kamal et al. 2020; Xiao et al. 2021). First, support vector machines have great generalization and nonlinear function approximation capabilities. Thus, Yang et al. (2008) used SVM to analyze and forecast freight rates, such as those of the China Coastal Bulk Freight Index (CCBFI), China Containerized



Freight Index (CCFI), and Baltic Freight Index (BFI). In addition, Guan et al. (2016) developed a forecasting model based on SVM and produced detailed projections using the dry bulk market's BSI data (Baltic Supramax Index).

In addition, BDI-related prediction research frequently uses artificial neural network models. One study first proposed a decomposition technique for BDI data and then used neural networks for prediction (Zeng et al. 2016). Another study developed three artificial neural network forecasting techniques using historical BDI data and global crude oil prices. The findings demonstrated that while the three neural network approaches perform similarly, ANN's prediction effect is superior (Şahin et al. 2018). Other studies have used wavelet neural networks, deep neural networks, and fuzzy neural networks to analyze and forecast the BDI in greater detail (Leonov and Nikolov 2012; Chou and Lin 2019; Kamal et al. 2020). These studies have increased the popularity of neural networks in researching dry bulk freight rate forecasting.

Zhang et al. (2018) compared the artificial neural network model with the econometric model. Their findings demonstrate that utilizing neural networks to predict the BDI is more accurate than conventional techniques. It is easier for neural network-based predictive models to handle complex nonlinear interactions and give accurate estimates in this area. Determining the model structure of AI algorithms, however, is challenging. This is susceptible to overtraining or undertraining, both of which can result in deficiencies, such as the sensitivity to initial values causing it to fall into local minima (Zhang et al. 2019). Furthermore, neural networks are particularly vulnerable to undertraining and overtraining. This vulnerability may result in shortcomings, such as the issue of becoming stuck in local minima due to starting value sensitivity (Zhang et al. 2018).

Consequently, several authors have discussed novel approaches to forecasting. For instance, a number of them believe that the fuzzy-Delphi adjustment method enhances the precision of the statistical forecast of the dry bulk shipping index, and it has produced positive results (Duru et al. 2012). In addition, deep learning is a potential technique for reliable BDI prediction (Kamal et al. 2020). Among the deep learning methods, researchers consider the *long short-term memory model* a practical approach for dealing with dry bulk cargo time series problems. However, this method has not received much attention in BDI forecasting research (Xiao et al. 2021). To its credit, the AI-based strategy has provided many superior outcomes compared to conventional econometric and statistical models. Nevertheless, the results of the current research on BDI forecasting are diverse. However, we still do not comprehend the complementarity, benefits, and drawbacks of various prediction algorithms and more research is required into deep learning or integration models.

2.3 Integration models

The BDI time series is typically volatile and nonstationary. As a result, the 'skills' of forecasting tools, including econometric models and AI-based approaches, have limitations. In addition, it has been shown that we cannot make statistically valid predictions using only these techniques (Kamal et al. 2020) and newer research is looking



into using integrated approaches to predict the BDI and improve prediction accuracy (Zhang et al. 2019). Indeed, various disciplines have widely employed *ensemble learning* to enhance model performance by increasing the prediction capacity of a single model (Xiao et al. 2021). Research has suggested combining noise-reduction methods with AI-based algorithms as a standard method for constructing ensemble models (Zhang et al. 2019). For this reason, authors employ a wavelet transform to remove noise from BDI data (Han et al. 2014). Researchers also use empirical mode decomposition (EMD) and ANN to create a separate integrated model for BDI prediction, achieving good accuracy (Zeng et al. 2016).

In another approach, Uyar et al. (2016) used genetic algorithms with a well-trained recurrent neural network to enhance the precision of long-term BDI forecasting. In addition, Zhang et al. (2019) predicted the BDI using a method that combined a dynamic fluctuation network (DFN) with AI. The authors found that the DFN-AI model had higher accuracy and trend-matching rate than its AI-based counterpart. Further, fusing several independent models into a single ensemble model can enhance the prediction accuracy of the BDI. In addition, Kamal et al. (2020) employed a deep integrated recurrent network consisting of a recurrent neural network (RNN), long short-term memory (LSTM), and gated rectifier unit neural network (GRU) to enhance predictive performance. The results of the ensemble models mentioned above indicate that hybrid models are superior to single AI models in time series forecasting. In reality, the hybrid forecasting approach combines the characteristics of each individual model and allows the respective benefits to compensate for the shortcomings of other methods (Zhang et al. 2019). Consequently, we predict that integrated forecasting models will become a focal point in future maritime freight market analyses. However, the computational requirements of hybrid methods are intricate and call for further innovative research.

3 Method

3.1 Data description

From January 6, 1995, to September 16, 2022, daily pricing data of BDI were obtained from Clarksons. Figure 1 displays the trend. The series does not appear to be stationary. The data exhibit significant fluctuations, high frequency, and distinctive characteristics, and the graph reveals major turning points at 2003, 2005, 2008, and 2021. Data points show substantial fluctuations at various time intervals. From 1999 to 2003, the BDI varied somewhat. Then, beginning in 2003, market volatility intensified. In 2008, the BDI reached a record high of 11,793. However, following the financial crisis of 2008–2009, a flood of newbuildings entered the market resulting in an excess supply of tonnage. In conjunction with a decline in demand, this caused the Baltic Dry Index (BDI) to plummet to one of its lowest levels (Oomen 2012), falling dramatically to 700 index points. Since then, the BDI has fluctuated continuously within a limited range. In February 2016, the index plummeted again to a 30-year low of 290. The index saw a brief recovery during the summer of the same year but subsequently proceeded to decline due to an oversupply of tonnage and a slower growth in commodities demand.



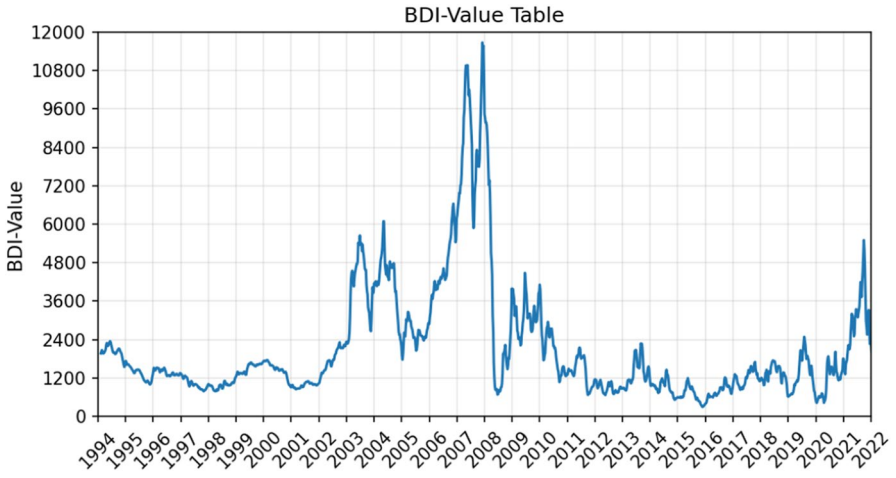


Fig. 1 Baltic Dry Index (BDI raw data)

As of 2019, the outbreak of the COVID-19 pandemic had a significant impact on shipping, resulting in substantial variations in BDI (Notteboom and Haralambides 2020; Cullinane and Haralambides 2021).

Table 1 presents the descriptive statistics of the data, including mean, standard deviation, skewness, kurtosis, and coefficient of variation. The skewness and kurtosis of the BDI series and its rate-of-change series are greater than zero. Furthermore, the standard deviation is high at 86.65% of the mean, indicating a high level of volatility in the freight market. In addition, the distribution of the series appears to be right-skewed, with evidence of fine kurtosis.

We assessed the BDI series using the Jarque–Bera test (Cromwell et al. 1994) and tested autocorrelation using Ljung and Box's (1978) test. Finally, we determined stationarity using the Augmented Dickey-Fuller (ADF) test (Fuller 2009). Based on these results, the BDI series exhibits non-normality and autocorrelation.

3.2 Model

We predict the BDI using the CNN-BiLSTM-AM ensemble model, which includes a convolutional neural network, a bi-directional long short-term memory, and an attention mechanism. Using CNN, one extracts characteristics from the data inputs. The BiLSTM then uses these features, extracted by the CNN from the data input, for BDI prediction. In addition, the AM captures the effect of significant past temporal feature states, to increase the accuracy of predictions.

Table 1 Descriptive statistics

Mean value	Standard deviation	Skewness	Kurtosis	Variation
2116.47	1833.95	2.45	6.86	0.87



3.2.1 Convolutional neural networks (CNN)

Convolutional neural networks are one of the most prominent deep learning algorithms. CNN is a feed-forward neural network with convolution calculation and a deep structure, highly suited to processing one-dimensional time domain sequences and picture data. Compared to other neural networks, CNN can efficiently and rapidly learn raw data and representations. Therefore, authors use it in various applications, including computer vision and natural language processing (Gu et al. 2018). The data convoluted in this study is 1D time series data, and 1D-CNN can extract the features of the data. The CNN model structure mainly includes input, convolution, pooling, fully connected, and output layers. We use a single convolutional layer to reduce data dimensionality and extract serial information. Figure 2 shows the model's structure.

Assuming a time series $(x_{t_1}, x_{t_2}, \ldots x_{t_k})$ of length k, after the convolution operation of n convolution kernels with stride = 1, we can obtain the output vector sequence $(o_1, o_2, \ldots o_{k-2})$. The calculation process is:

$$c_{i,j} = \sum_{p=1}^{m} w_{i,p} x_{t_{i+p-1}}$$

$$o_i = [c_{1,i}, c_{2,i}, \ldots, c_{n,i}]$$

3.2.2 Bi-long short-term memory network (BiLSTM)

The long short-term memory network is a deep learning model widely used in time series data and it is an improved version of the short-term memory network. The basic concepts in LSTM networks are "gate" and "unit" states, which are "forget gate," "input gate," "unit state," "output gate," and "hidden state," respectively. The

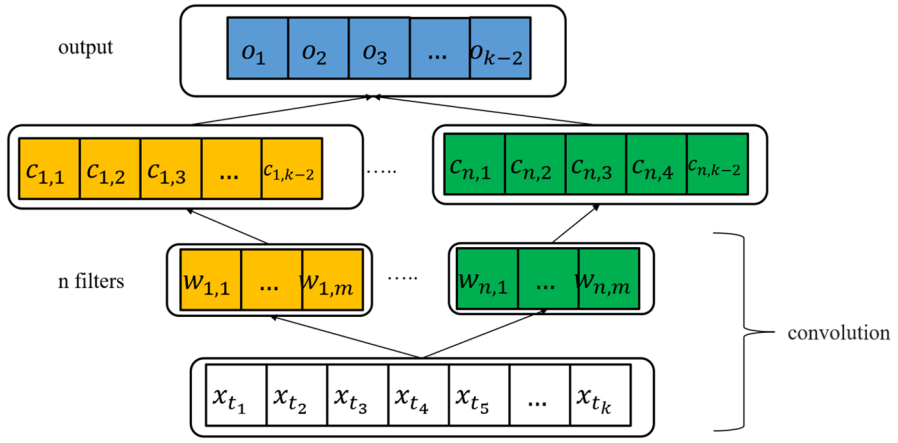


Fig. 2 Convolutional neural network structure diagram

LSTM is a kind of temporal recurrent neural network (RNN). It is suitable for processing and predicting important events with long intervals and time series delays and it can solve well the long-term dependence problem in simple RNNs (Fischer and Krauss 2018). Figure 3 shows the LSTM's cell schematic diagram. The LSTM contains three gates: forget, input, and output. C_{t-1} is the cell state at the previous moment, h_{t-1} is the final output value of the LSTM neuron unit at the previous moment, x_t is the input at the current moment, σ is the Sigmoid activation function, and tanh is also the activation function.

The LSTM calculation process is as follows:

f_t is the output of the forget gate at the current moment:

$$f_t = \sigma(W_{\mathrm{f}} \cdot [h_{t-1}, x_t] + b_{\mathrm{f}})$$

i_t is the output of the input gate at the current moment:

$$i_t = \sigma(W_{\mathrm{i}} \cdot [h_{t-1}, x_t] + b_{\mathrm{i}})$$

$\tilde{C}_t$ is the current candidate cell state:

$$\tilde{C}_t = tanh(W_{\mathrm{C}} \cdot [h_{t-1}, x_t] + b_{\mathrm{C}})$$

o_t is the output value of the output gate:

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t$$

C_t is the cell state at the current moment:

$$o_t = \sigma(W_{\mathrm{o}} \cdot [h_{t-1}, x_t] + b_{\mathrm{o}})$$

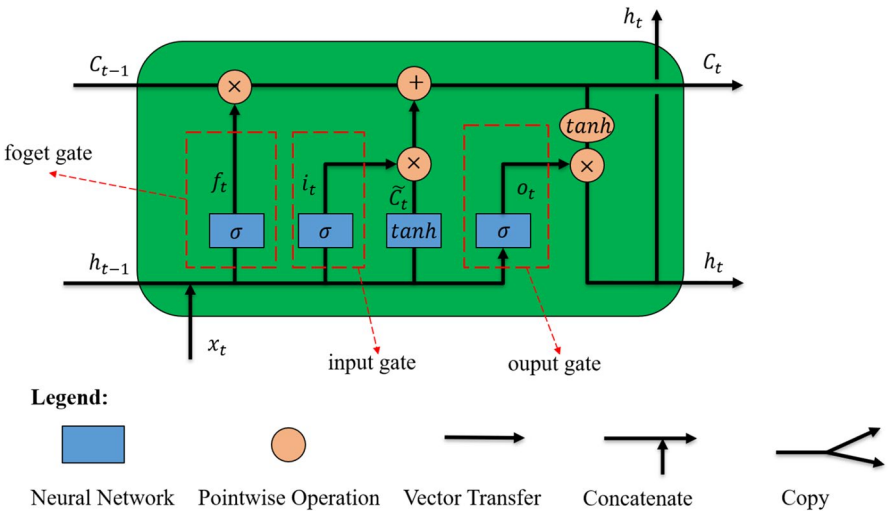


Fig. 3 Long short-term memory network structure diagram

and h_t is the output at the current moment:

$$h_t = o_t \times \tanh(C_t)$$

W_f, W_i, W_C, and W_o are the weights of the forget gate, input gate, candidate input gate, and output gate, respectively. In addition, b_f, b_i, b_C, and b_o are the biases of the forget gate, input gate, candidate input gate, and output gate, respectively. As the name suggests, the bi-directional long short-term memory network can memorize information in both directions. Its basic composition is the LSTM unit. The output vector of forward LSTM at time $\overrightarrow{h_t}$ and the output vector of the backward LSTM $\overleftarrow{h_t}$, after connecting left and right, becomes the output H_t of BILSTM. Figure 4 shows the forward LSTM combined with the backward LSTM to form BiLSTM (Li et al. 2018).

The output of the hidden layer of the BiLSTM cell at time t is a combination of the output $\overrightarrow{h_t}$ of the forward LSTM and the output $\overleftarrow{h_t}$ of the backward LSTM:

$$y_t = [\overrightarrow{h_t}, \overleftarrow{h_t}]$$

3.2.3 Attention mechanism (AM)

The AM principle derives from the variation in human visual attention, which one can use to allocate different feature information resources. In recent years, combining attention models and deep learning has produced remarkable results in various domains like stock market predictions (Lu et al. 2021). Through the attention probability distribution, we can determine the importance of the prediction's different features in the prediction problem and assign different weights to reduce the error between the predicted value and the actual value (Niu et al. 2021).

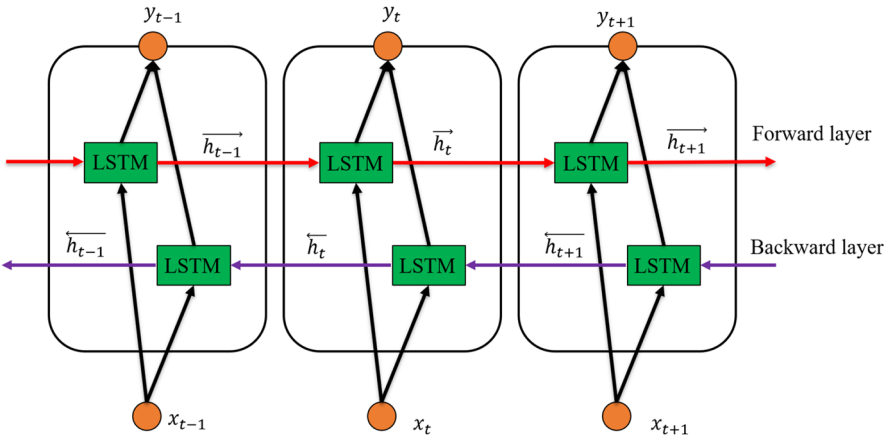


Fig. 4 Bi-long short-term memory network structure diagram

3.3 CNN-BiLSTM-AM model

Figure 5 shows the CNN-BiLSTM-AM model framework. We use keras to build the model, and the CNN and BiLSTM layers have specific APIs in the keras framework, which one can directly combine and call.

The AM layer is custom-built, usually in the following three steps:

(1) Compute the similarity between the output value and input characteristic:

$$s_t = \tanh(W_h h_t + b_h)$$

where W_h is the weight of AM, b_h is the bias of AM, and h_t is the input vector, the BiLSTM layer's output vector.

(2) Normalize the results in the first step, and then use the softmax function to convert the attention score to obtain the attention weight coefficient:

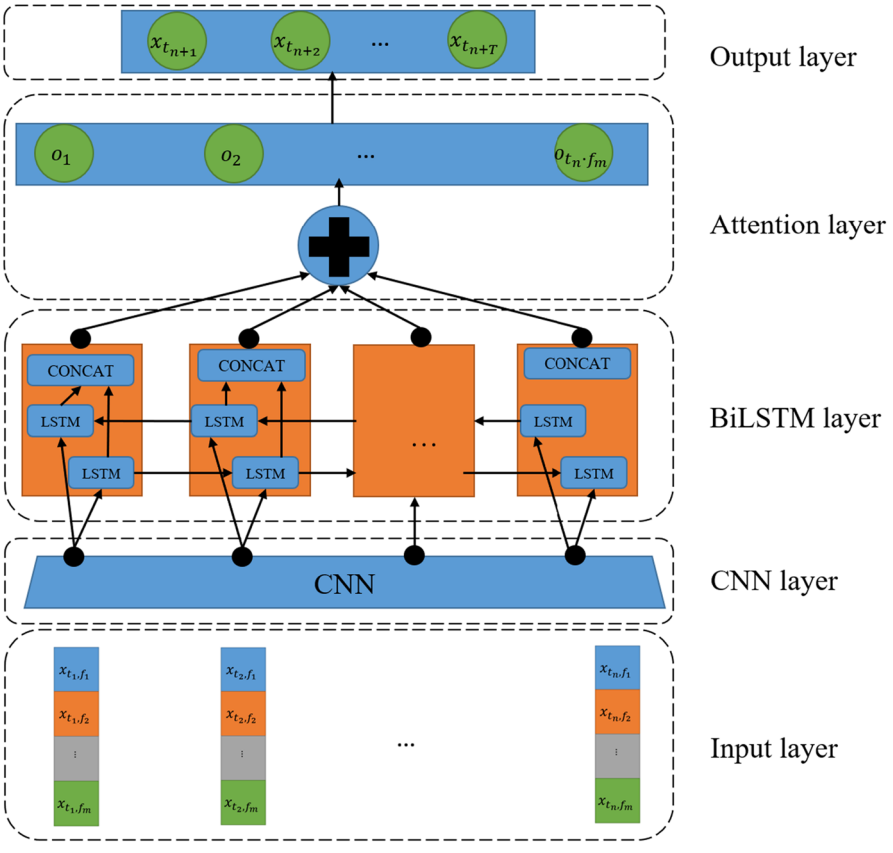


Fig. 5 CNN-BiLSTM-AM model framework

$$a_t = \frac{\exp(s_t^{\mathrm{T}} v)}{\sum \exp(s_t^{\mathrm{T}} v)}$$

v is the attention value, the same as the input layer vector in this study.

(3) According to the weight coefficient, the input values are weighted and summed to obtain the final output:

$$s = \sum_t a_t h_t$$

Specific information on the final model parameters can be found in Table 3 in Appendix.

4 Results analysis

4.1 Data processing

To use the CNN-BiLSTM-AM integrated model we need to normalize the dataset $\mathrm{BDI} = \{\mathrm{BDI}_i | i = 1, 2 \ldots n\}$:

$$\mathrm{BDI}_i^{'} = \frac{\mathrm{BDI}_i - \min(\mathrm{BDI})}{\max(\mathrm{BDI}) - \min(\mathrm{BDI})}$$

In addition, because the data must enter the LSTM neural network for calculation, we use the first n points of BDI to forecast the BDI at the following m points. Therefore, we must partition the dataset according to the above. As depicted in Fig. 6, the initial n data points predict the following m ones, generating many data pairs in succession. We then divide the dataset 7:3 to create the neural network's training and test sets. Finally, we use the training set to train the model's predictive capacity and the test set to evaluate the trained model's performance.

4.2 Estimation results

We have compared in depth the prediction results of a single model versus an ensemble one. The result graphs of the various models follow, with the first graph representing the performance of the training set and the second graph illustrating the test set's performance (Figs. 7, 8, 9, 10, 11, and 12).

Epochs = 1,000, Loss Function: MSE, Optimizer: Adam, Batch size: 256, Learning rate: 0.001.

4.3 Model evaluation

We employ MSE and R^2 as the evaluation measures. The MSE and R^2 formulas are as follows:

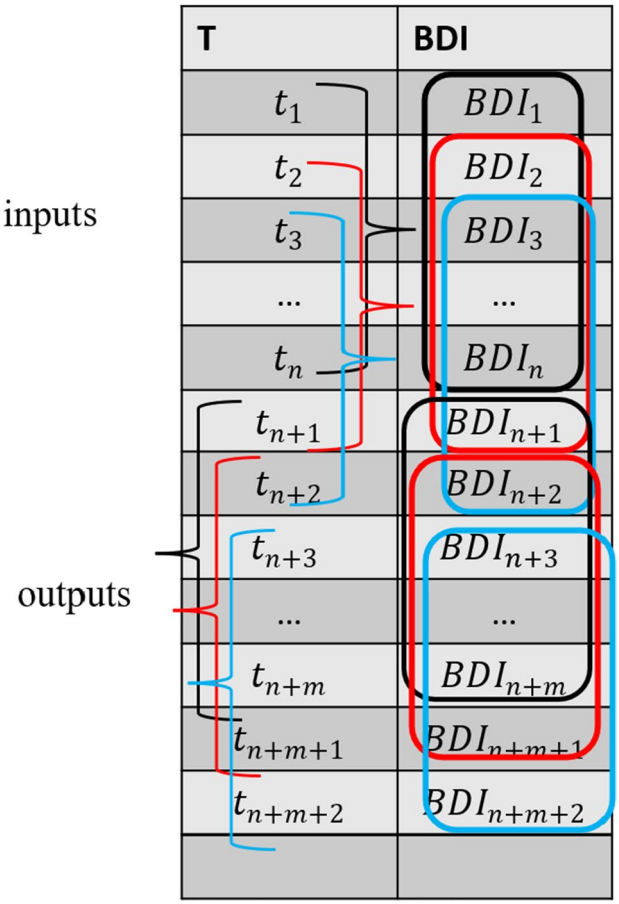


Fig. 6 Dataset partition

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(\overline{y_i} - \hat{y}_i\right)^2}$$

where y_i is the true value; $\hat{y}_i$ is the predicted value; $\overline{y_i}$ is the average value.

Following 100 simulation rounds under each model, we obtain the mean MSE and R^2 values for each model on the training and test sets, presented in Table 2. The comparison in Table 2 shows that the CNN-BiLSTM-AM integrated model has the best performance, with an R^2 of 96.9%.

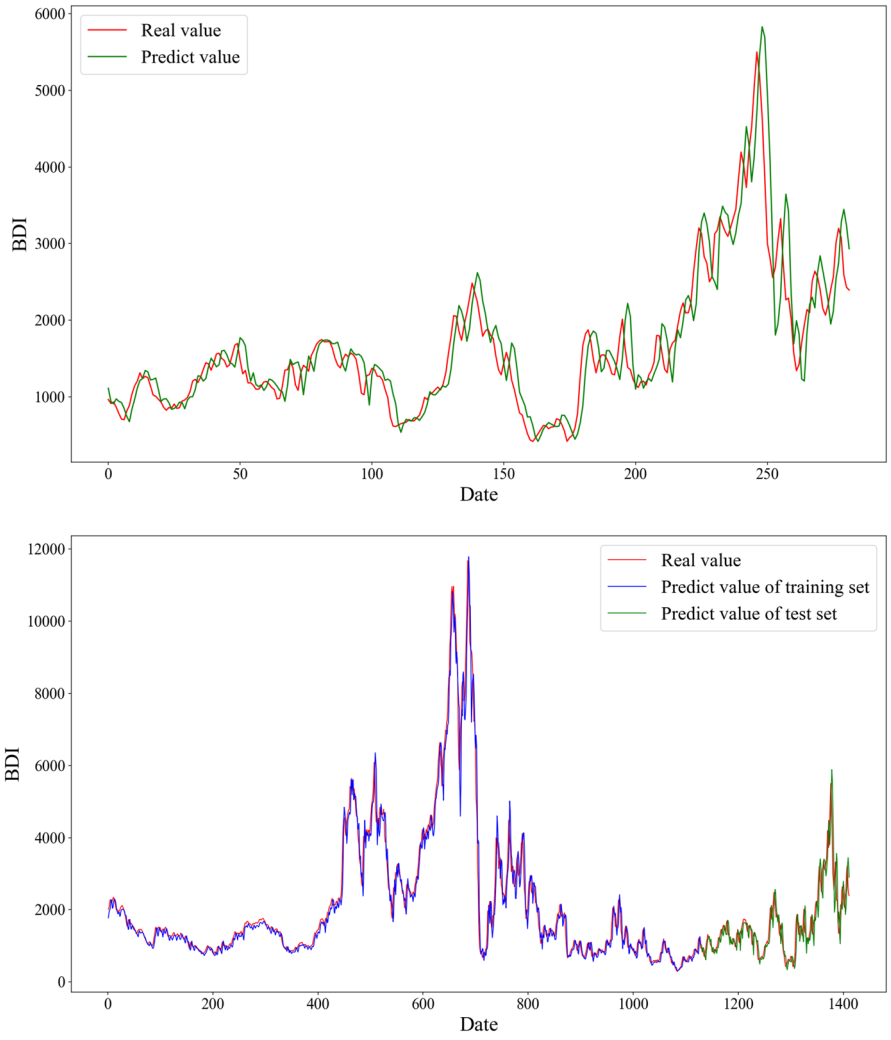


Fig. 7 CNN model results

5 Discussion and conclusion

BDI is a major indicator of the shipping sector and, often, of international trade and the worldwide economy (Wang et al. 2015). In reality, forecasting BDI trends allows operators and investors to manage market trends and avert risks. This guides decision-making in other industries and manufacturers in the global economy (Papapostolou et al. 2016). In contrast to typical time series data, the BDI index exhibits volatility, non-stationarity, and complexity. Indeed, its prediction is quite challenging. In light of these features of BDI, we propose an ensemble model based on deep learning to forecast the trend of the BDI. We offer a



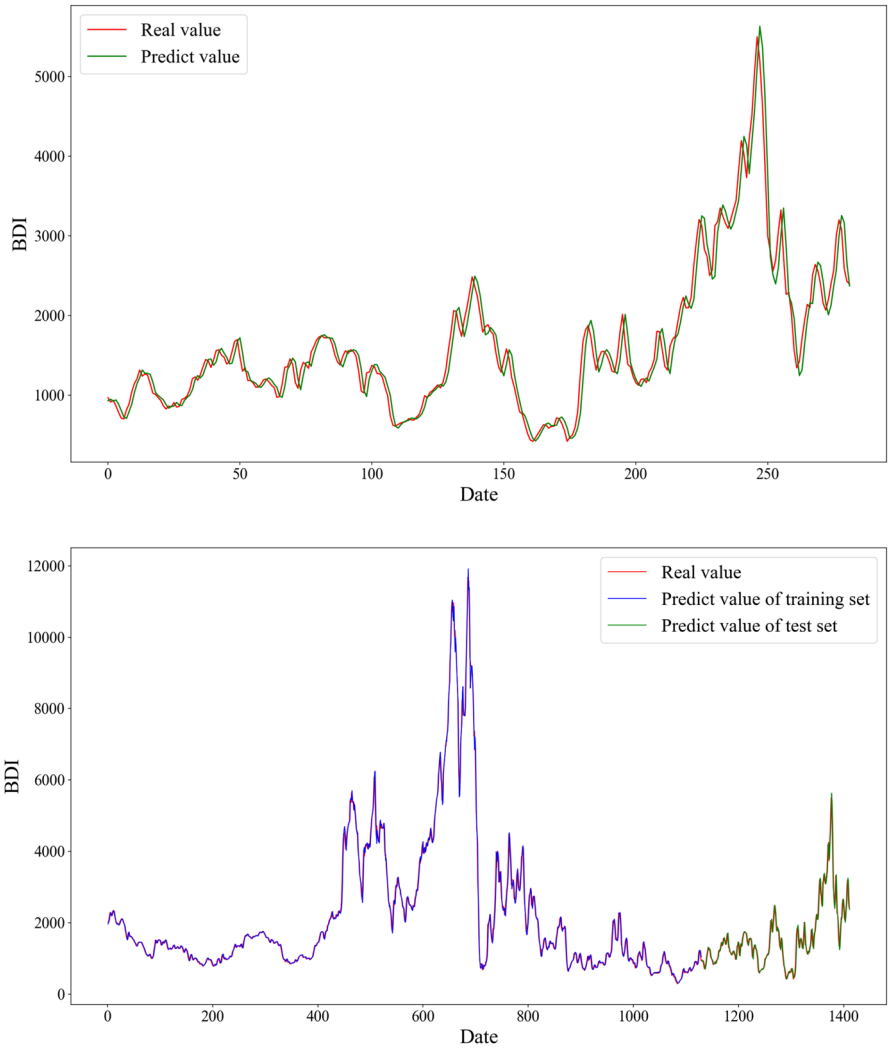


Fig. 8 LSTM model results

deep learning ensemble method that incorporates CNN, BiLSTM, and AM into a set of deep learning ensemble models, aiming to increase the accuracy of predictions. Using 6,896 observations, we have evaluated the prediction power of the suggested model. The study captures the hidden nonlinear features of BDI using machine learning techniques backed by artificial neural networks. Specifically, with CNN's assistance, we caught hidden nonlinearities to predict BDI using feature data from BiLSTM and deep learning. The AM captures the effect of past significant temporal feature states on the BDI, to increase the accuracy of predictions. Our findings indicate that the CNN-BiLSTM-AM ensemble model outperforms a single machine learning one. Moreover, the model demonstrates

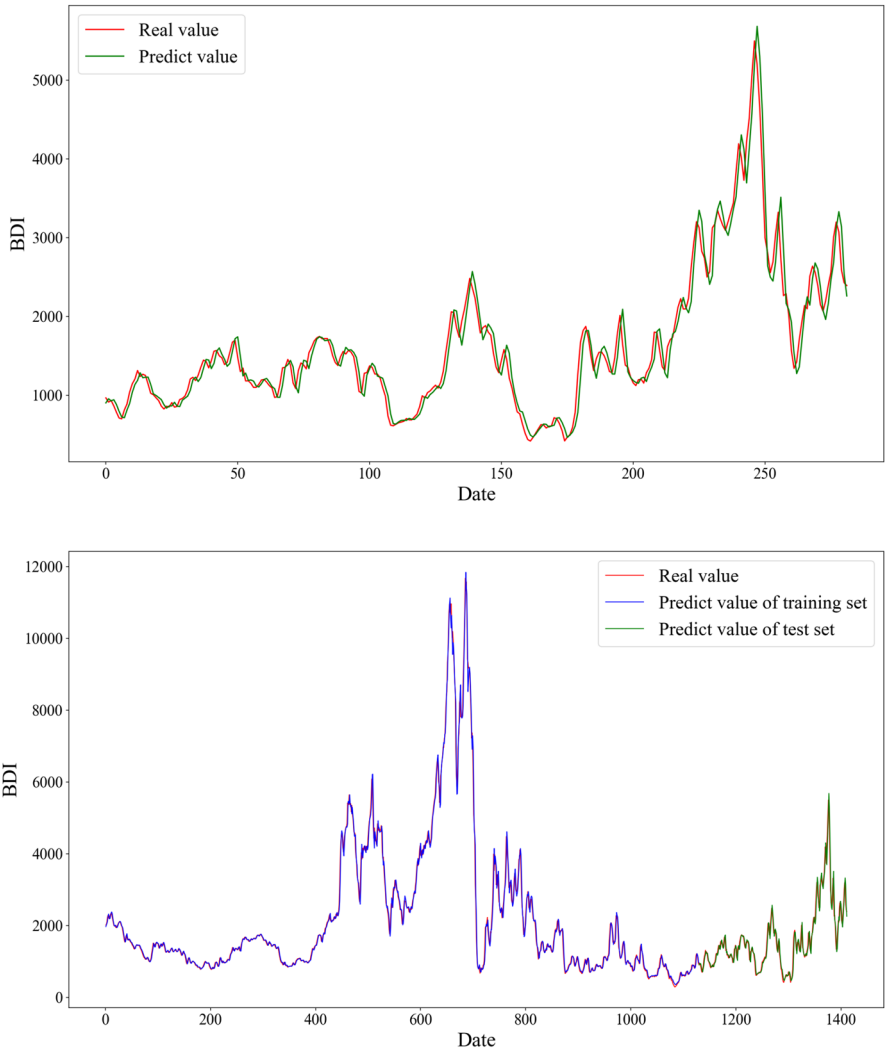


Fig. 9 BiLSTM model results

superior performance compared to ensemble models incorporating other models ($R^2 = 96.9\%$).

This paper makes multiple contributions. First, the paper complements existing machine learning approaches in predictive research. Considering the nonlinearity and high volatility of the BDI, our study's prediction findings indicate that the CNN-BiLISM-AM deep integration model can forecast the BDI's change trend. Deep learning ensemble approaches can efficiently learn meaningful information from features by extracting multi-layer representations of the features, hence attaining higher prediction accuracy than machine learning methods that use shallow features



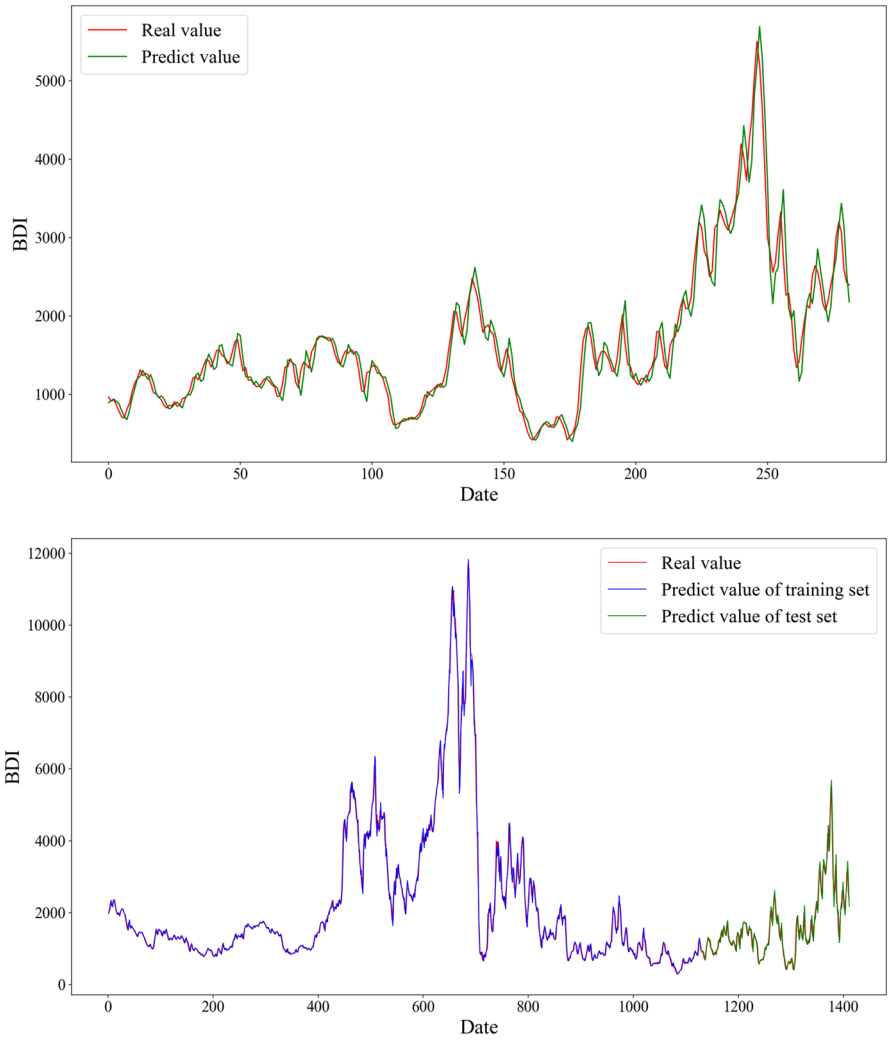


Fig. 10 CNN-LSTM model results

for prediction. Similar to Kamal et al.'s (2020) BDI prediction study, our results support the superior performance of LSTM-based deep ensemble models over single deep learning methods. In addition, compared to Makridakis et al. (2020) study, our model captures the variations of key nodes of the BDI and delivers more accurate predictions than prior studies. This finding further demonstrates the excellent implementation of AM in forecasting.

This work also contributes to shipping research. We primarily viewed the BDI as a forward-looking indicator of economic activity, investigated its time path, and presented a deep learning-based integrated model. From BDI data, we attempted to deduce shipping industry trends. Focusing on the value of the data has a distinct

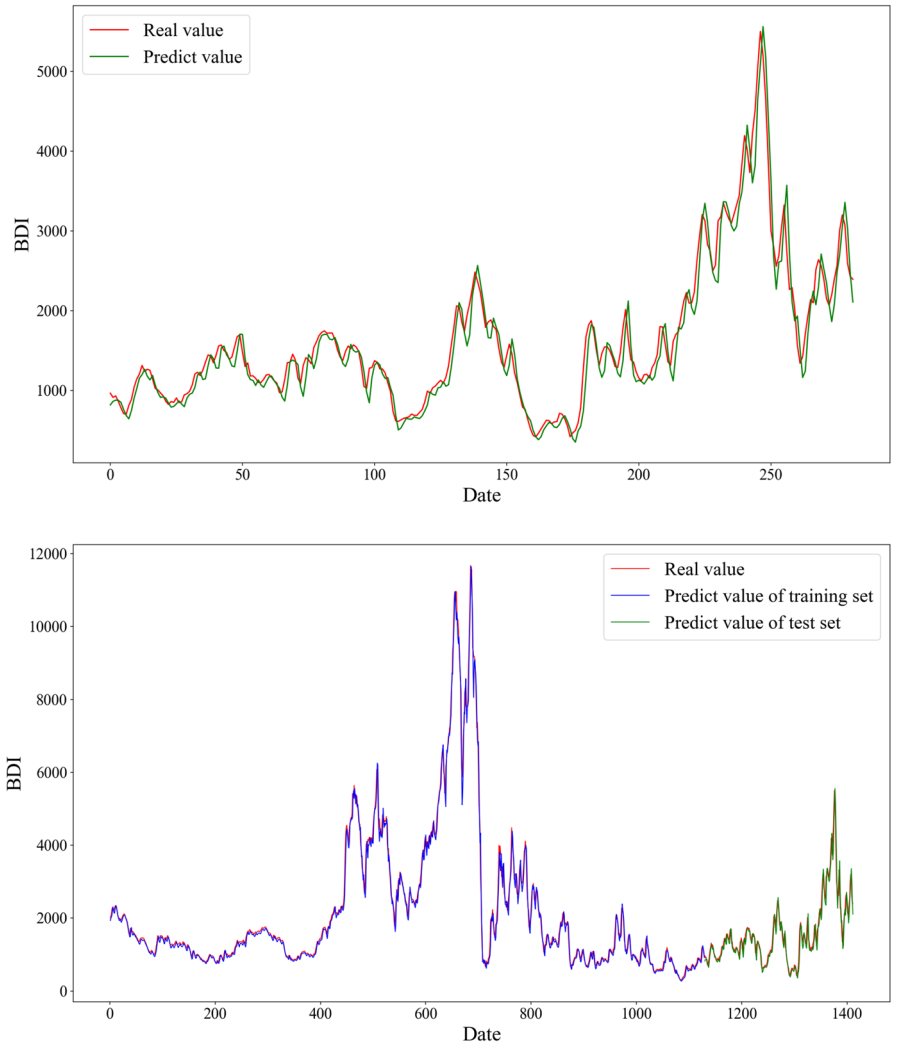


Fig. 11 CNN-BiLSTM model results

attraction in the age of intelligence, i.e., utilizing a data-driven model selection procedure. The technique is based entirely on the statistical features of the data, seeking to identify the correct univariate time series model employing heteroskedasticity, departure from normality, and long memory data characteristics. We concur with the findings of Katris and Kavussanos (2021) and extend the research on BDI univariate prediction.

The study concludes by offering certain management insights. Our Maritime Transport Forecast Model serves as a valuable tool in providing insights to the global economic landscape. Thus, we can deduce economic activity trends from shipping statistics. First, shipowners can make business decisions using

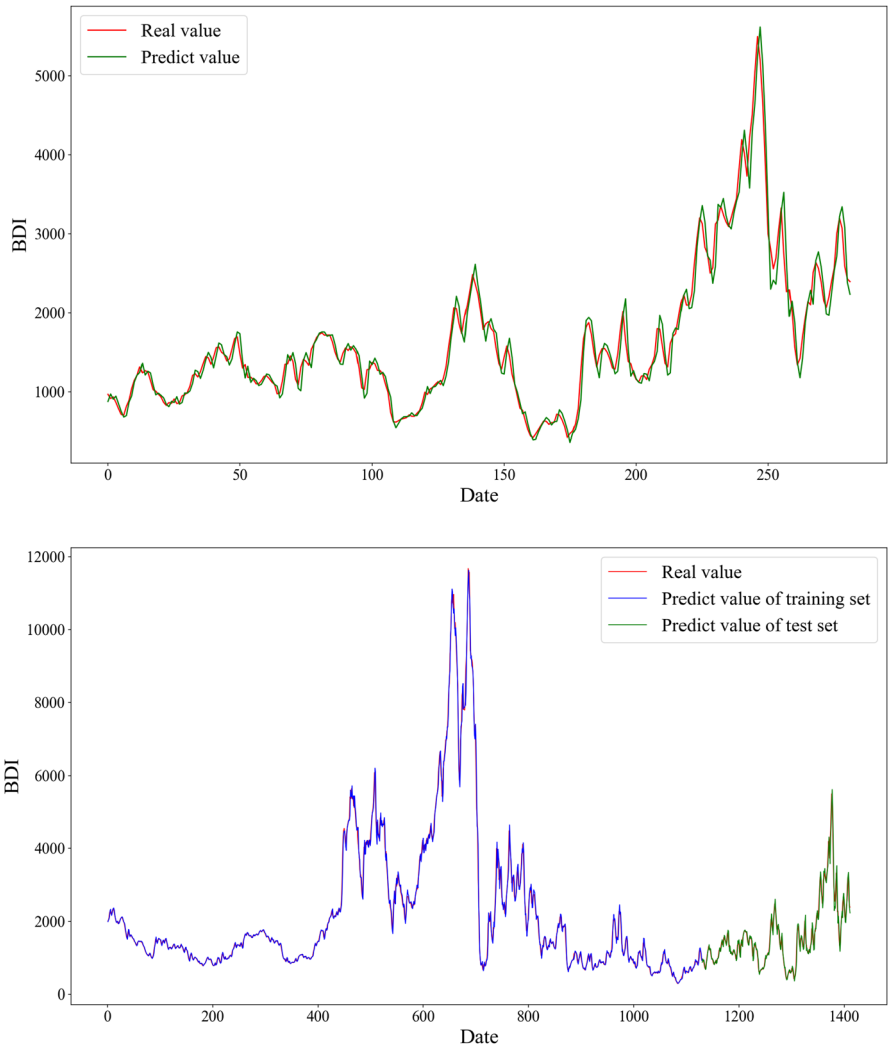


Fig. 12 CNN-BiLSTM-AM model results

Table 2 Model performance comparison

Model	Training set	Test set	Training set R^2	Test set R^2
CNN	8.7146e−04	0.0011	97.3	81.8
LSTM	2.6141e−04	3.3773e−04	99.1	94.6
BiLSTM	2.3394e−04	3.4468e−04	99.2	94.3
CNN-LSTM	2.7196e−04	3.2752e−04	99.1	94.7
CNN-BiLSTM	2.9995e−04	3.2857e−04	99.0	94.6
CNN-BiLSTM-AM	1.4476e−04	2.1469e−04	99.7	96.9

short-term information about seasonal swings in the bulk cargo industry. If the expected BDI is low, the ship can go to a dry dock for maintenance and repairs or be laid-up or sold. When the BDI is expected to be high, shipowners, often within alliances, can coordinate better operations and investments for profit optimization. When the BDI trend rises, shipping corporations can decide to increase the speed of their ships to boost productivity. Moreover, investors can consider purchasing, holding, or selling shipping company stocks when considering the anticipated volatility of the BDI.

This study is not void of drawbacks. First, it is suggested that focusing on the value of BDI data itself may be an intriguing topic. However, the study focuses solely on the changes in the BDI time series, a single dataset. Second, we combine a limited number of machine learning models into model ensembles. As a result, it is easy to overlook more combinations of machine learning models with better performance. Third, the duration predictions may not be optimal. Future research models should enhance prediction lengths. Therefore, we advise including more market-influencing factors and carrying out multi-factor projection research. In addition, we propose discovering more efficient deep learning techniques to improve prediction cycle and accuracy.

Appendix

Table 3 illustrates the specific information of the final model parameters.

Table 3 Final model parameters

Layer (Type)	Output shape	Param#
Input_1 (InputLayer)	[(None, 14, 1)]	0
Conv1d (conv1D)	(None, 14, 64)	256
Dropout (Dropout)	(None, 14, 64)	0
Bi-directional (Bi-directional)	(None, 14, 128)	66,048
Dropout_1 (Dropout)	(None, 14, 128)	0
Dense (Dense)	(None, 14, 128)	16,512
Attention_vec (Permute)	(None, 14, 128)	0
Multiply (Multiply)	(None, 14, 128)	0
Flatten (Flatten)	(None, 14, 128)	0
Dense_1 (Dense)	(None, 1)	1793
Total params: 84,609 Trainable params: 84,609 Non-trainable params: 0		



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